

Product Requirements Document (PRD)

1. Product Overview

Product Name

Environmental Intelligence Drone (EID) Web Platform

Product Description

The Environmental Intelligence Drone Web Platform is a web-based environmental monitoring and decision-support system designed to receive, process, visualize, and analyze data collected by the Environmental Intelligence Drone (EID).

The platform connects the drone's **ESP32-based sensor system, GPS, and LoRa communication module** with a centralized dashboard. It enables users to monitor environmental conditions in real time, visualize GPS-tagged pollution measurements on an interactive map, analyze historical trends, view AI-based pollution-risk predictions, and receive early-warning notifications.

The platform is intended to complement fixed environmental monitoring infrastructure by providing mobile and spatially distributed environmental intelligence.

Core Product Flow

Drone → Sensors → ESP32 → GPS Tagging → LoRa → Ground Station → Web Platform → Pollution Mapping → AI Prediction → Risk Analysis → Early Warning

2. Problem Statement

Conventional environmental monitoring systems often rely on fixed monitoring stations. These systems provide measurements at specific locations but may have limited ability to capture the spatial movement of pollutants across large or changing areas.

The EID Web Platform addresses this gap by transforming mobile drone-collected measurements into:

- Real-time environmental information
- GPS-based pollution maps
- Historical environmental trends
- Pollution-risk analysis
- AI-based prediction
- Early-warning notifications

The platform should help users answer four key questions:

1. **What is happening?**
 2. **Where is it happening?**
 3. **How is it changing?**
 4. **Where could the risk increase next?**
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3. Product Goals

Primary Goals

1. Display real-time environmental data collected by the drone.
 2. Visualize GPS-tagged sensor measurements on an interactive map.
 3. Generate dynamic pollution-intensity maps.
 4. Display current and historical environmental trends.
 5. Provide AI-based pollution-risk/trend prediction.
 6. Generate early-warning alerts for abnormal or potentially unsafe conditions.
 7. Display drone location and survey path.
 8. Provide a simple interface for environmental authorities and researchers.
 9. Maintain a historical environmental data record.
 10. Provide a scalable architecture for additional sensors and future multi-drone deployment.
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4. Target Users

Primary Users

Municipal Authorities

Use the platform to monitor environmental conditions across selected areas and identify pollution hotspots.

Pollution-Control Authorities

Use environmental measurements and spatial maps to support monitoring and investigation.

Landfill Operators

Monitor methane-related conditions and environmental variations around landfill areas.

Industrial Facilities

Monitor environmental conditions around industrial zones and identify abnormal readings.

Secondary Users

- Smart-city administrators
 - Disaster-management teams
 - Environmental researchers
 - Academic institutions
 - Project administrators
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5. Product Scope

In Scope

- User authentication
- Environmental dashboard
- Live drone status
- Live sensor readings
- GPS tracking
- Flight-path visualization
- Pollution heatmap
- Historical data visualization
- AI prediction visualization
- Risk classification
- Early-warning alerts
- Mission/area visualization
- Sensor data storage
- Drone telemetry
- Data export
- System settings
- Device/sensor status
- GitHub/project information page

Out of Scope for Initial Version

- Direct control of industrial equipment
- Automatic pollutant neutralization
- Medical diagnosis
- Certified regulatory environmental measurement
- Fully autonomous multi-drone coordination
- Public emergency dispatch
- Direct drone video streaming as a core feature
- Automated enforcement decisions

6. Core Features

6.1 Dashboard

The dashboard is the primary screen after login.

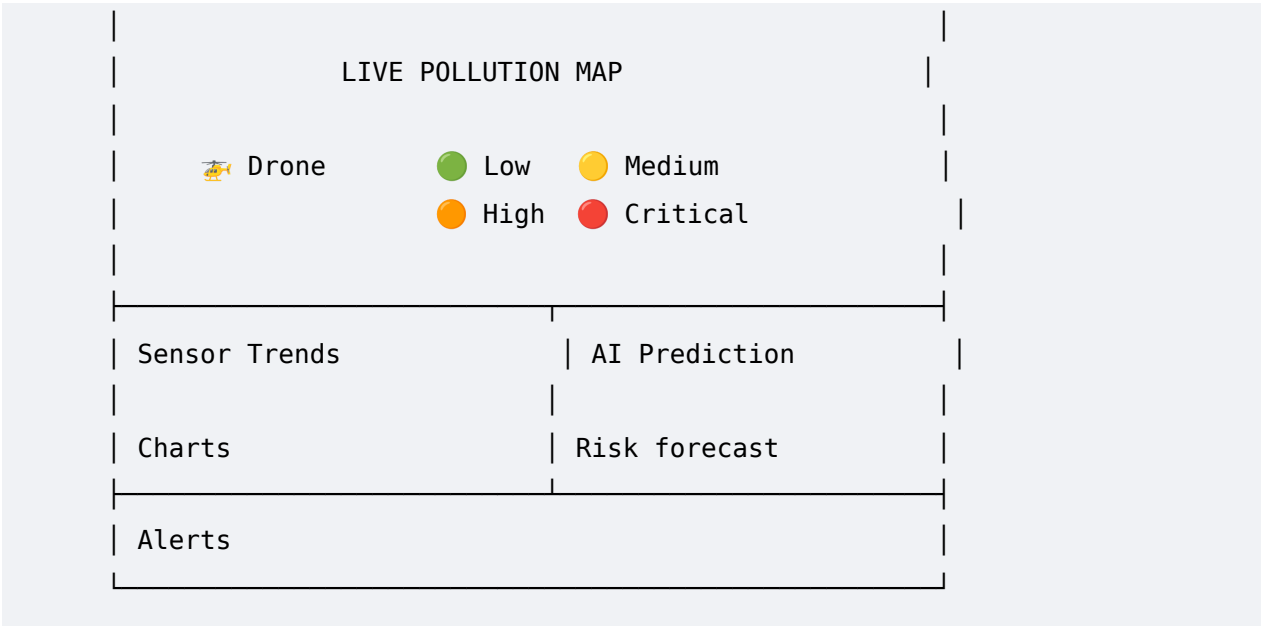
Dashboard should display

Top-level metrics:

- Current methane status
- Air-quality status
- Temperature
- Humidity
- Atmospheric pressure
- Wind speed
- Wind direction
- Drone battery
- GPS status
- LoRa communication status
- Current risk level

Example layout

Environmental Intelligence Dashboard			
Methane	Air Quality	Temp	Risk
NORMAL	MODERATE	31°C	LOW



7. Live Environmental Monitoring

The platform should receive sensor readings from the EID system.

Supported parameters

- Methane
- Air-quality sensor reading
- Temperature
- Humidity
- Atmospheric pressure
- Wind speed
- Wind direction

Each measurement should contain

Measurement ID
Drone ID
Timestamp
Latitude
Longitude
Methane value
Air-quality value
Temperature
Humidity
Pressure

Wind speed
Wind direction

Functional Requirement

FR-01: The platform shall display the latest available environmental sensor readings.

FR-02: The platform shall associate environmental readings with GPS coordinates and timestamps.

FR-03: The platform shall indicate when data has become stale or communication has been interrupted.

8. Live Drone Monitoring

The platform should display the current drone status.

Information

- Drone ID
- Connection status
- GPS status
- Latitude
- Longitude
- Altitude, if available
- Battery percentage
- Flight status
- Current mission
- Last data received
- LoRa connection status

Drone status

-  Online
 -  Limited Connection
 -  Offline
-

9. Interactive Pollution Map

The map is one of the most important features of the platform.

Map should display

- Drone's current position

- Drone flight path
- Measurement points
- Pollution heatmap
- Monitoring area
- Risk zones
- Predicted pollution zones

Map controls

Users should be able to:

- Zoom in/out
- Select monitoring area
- View measurement points
- Click a measurement point
- View sensor values
- Toggle heatmap
- Toggle predicted-risk layer
- Toggle flight path
- Select a time period

Measurement popup

When a user clicks a point:

Measurement #1042

Time:

14:32:15

Location:

17.xxxx, 78.xxxx

Methane:

Elevated

Air Quality:

Moderate

Temperature:

31°C

Humidity:

62%

Risk:

HIGH

10. Pollution Heatmap

The system should transform GPS-tagged environmental measurements into a spatial visualization.

Risk visualization

- 
- Low
 - Moderate
 - High
 - Critical

The map should clearly distinguish:

Measured Data

from

Predicted Risk

The platform should not represent interpolated or predicted values as direct sensor measurements.

11. Historical Analytics

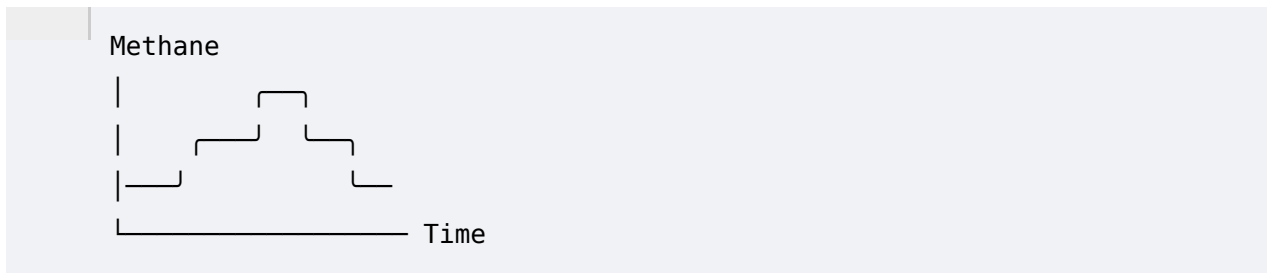
Users should be able to analyze environmental conditions over time.

Time filters

- Last hour
- Last 6 hours
- Last 24 hours
- Last 7 days
- Custom date range

Charts

Methane Trend



Air Quality Trend

Display changes over time.

Temperature/Humidity

Show environmental conditions.

Pollution Trend

Show changes in pollution-risk level.

12. AI Prediction Module

The AI module should provide environmental trend and risk predictions based on available sensor and environmental data.

Input Features

- Historical methane readings
- Air-quality readings
- Temperature
- Humidity
- Atmospheric pressure
- Wind speed
- Wind direction
- GPS coordinates
- Timestamp
- Historical measurements

AI output

The platform should display:

- Predicted pollution intensity

- Pollution-risk level
- Potential risk zone
- Predicted trend
- Confidence/uncertainty indicator when supported by the model

Important product rule

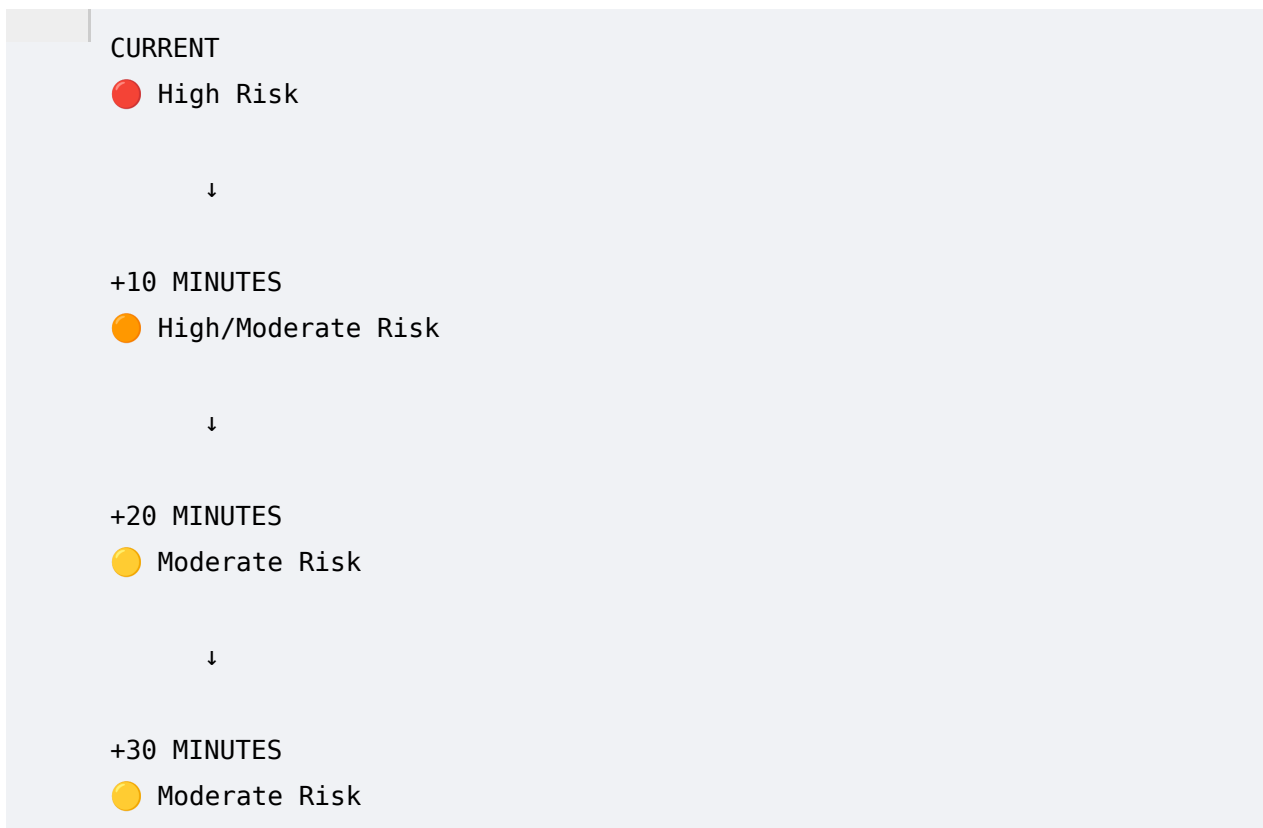
The platform must clearly label AI output as:

Predicted / Estimated Risk

and never present it as a confirmed sensor measurement.

13. AI Prediction Visualization

Example:



The map can show the predicted risk area as a separate layer.

14. Risk Assessment

The platform should combine current environmental readings and available prediction results to classify environmental risk.

Risk levels

Level	Meaning
 Normal	No significant abnormality detected
 Moderate	Increasing or unusual readings
 High	Significant environmental risk
 Critical	Severe or rapidly increasing risk

Risk factors

The risk engine can consider:

- Current sensor readings
- Historical trend
- Rate of change
- AI prediction
- Weather conditions
- Spatial concentration

Thresholds should be configurable by administrators and should not be presented as universal regulatory limits unless supported by the appropriate standards.

15. Early-Warning Alert System

The system should generate an alert when configured risk conditions are met.

Alert information

HIGH ENVIRONMENTAL RISK

Location:

17.xxxx, 78.xxxx

Parameter:

Methane

Status:

Elevated

Risk:

HIGH

Trend:

Increasing

Time:

14:35:42

Alert priorities

- Normal
- Information
- Moderate
- High
- Critical

Alert channels

Initial version:

- Dashboard notification

Future:

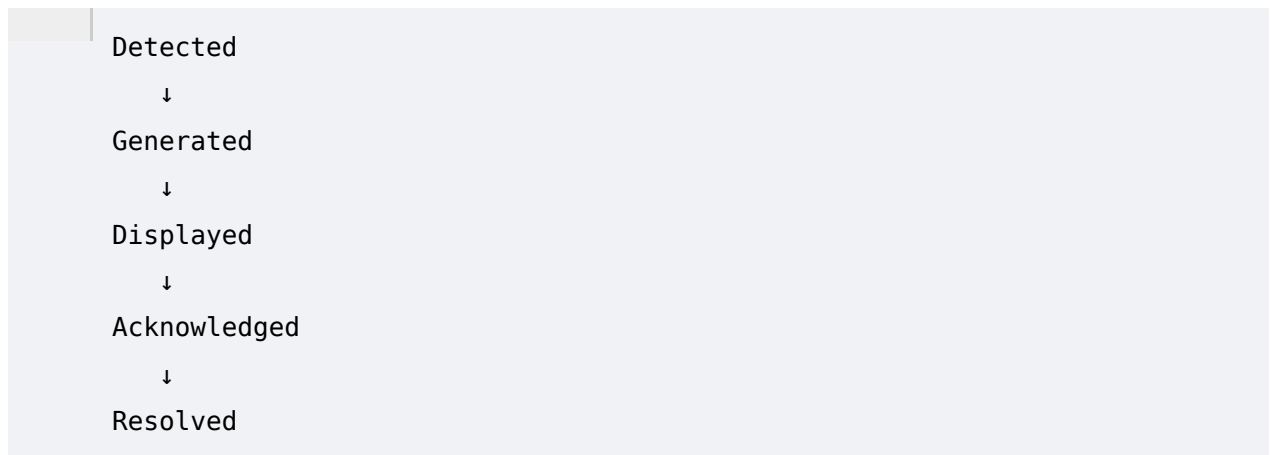
- Email
- SMS
- Mobile push notification

16. Alert Management

Users should be able to:

- View active alerts
- View historical alerts
- Filter alerts by severity
- Filter alerts by location
- Mark alerts as acknowledged
- View alert details
- Track alert status

Alert lifecycle



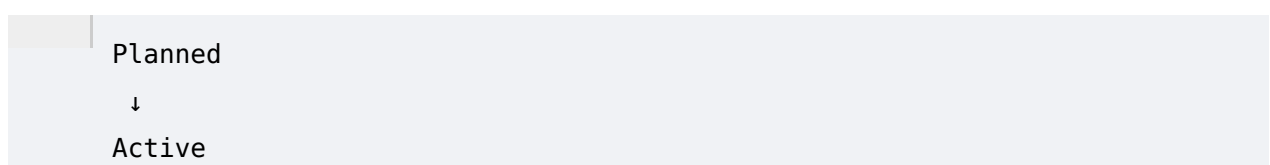
17. Mission Monitoring

The platform should allow users to view drone survey missions.

Mission information

- Mission ID
- Monitoring area
- Start time
- End time
- Drone ID
- Waypoints
- Flight path
- Number of measurements
- Mission status

Mission status



↓
Completed

18. Data Management

The platform should store environmental measurements for future analysis.

Database entities

Users

```
user_id
name
email
role
password_hash
created_at
```

Drones

```
drone_id
name
status
battery
gps_status
last_seen
```

Measurements

```
measurement_id
drone_id
timestamp
latitude
longitude
methane
air_quality
temperature
humidity
pressure
```

```
wind_speed
wind_direction
```

Missions

```
mission_id
drone_id
area
start_time
end_time
status
waypoints
```

Predictions

```
prediction_id
timestamp
location
predicted_risk
predicted_intensity
model_version
confidence
```

Alerts

```
alert_id
timestamp
location
severity
parameter
value
message
status
```

19. User Roles

Administrator

Can:

- Manage users
- Manage drones
- Configure thresholds
- View all missions
- View all data
- Manage system settings

Environmental Analyst

Can:

- View dashboard
- Analyze pollution maps
- View historical data
- View AI predictions
- Export reports
- Acknowledge alerts

Viewer

Can:

- View dashboard
- View maps
- View sensor readings
- View alerts

20. Reports and Data Export

Users should be able to generate environmental monitoring reports.

Report contents

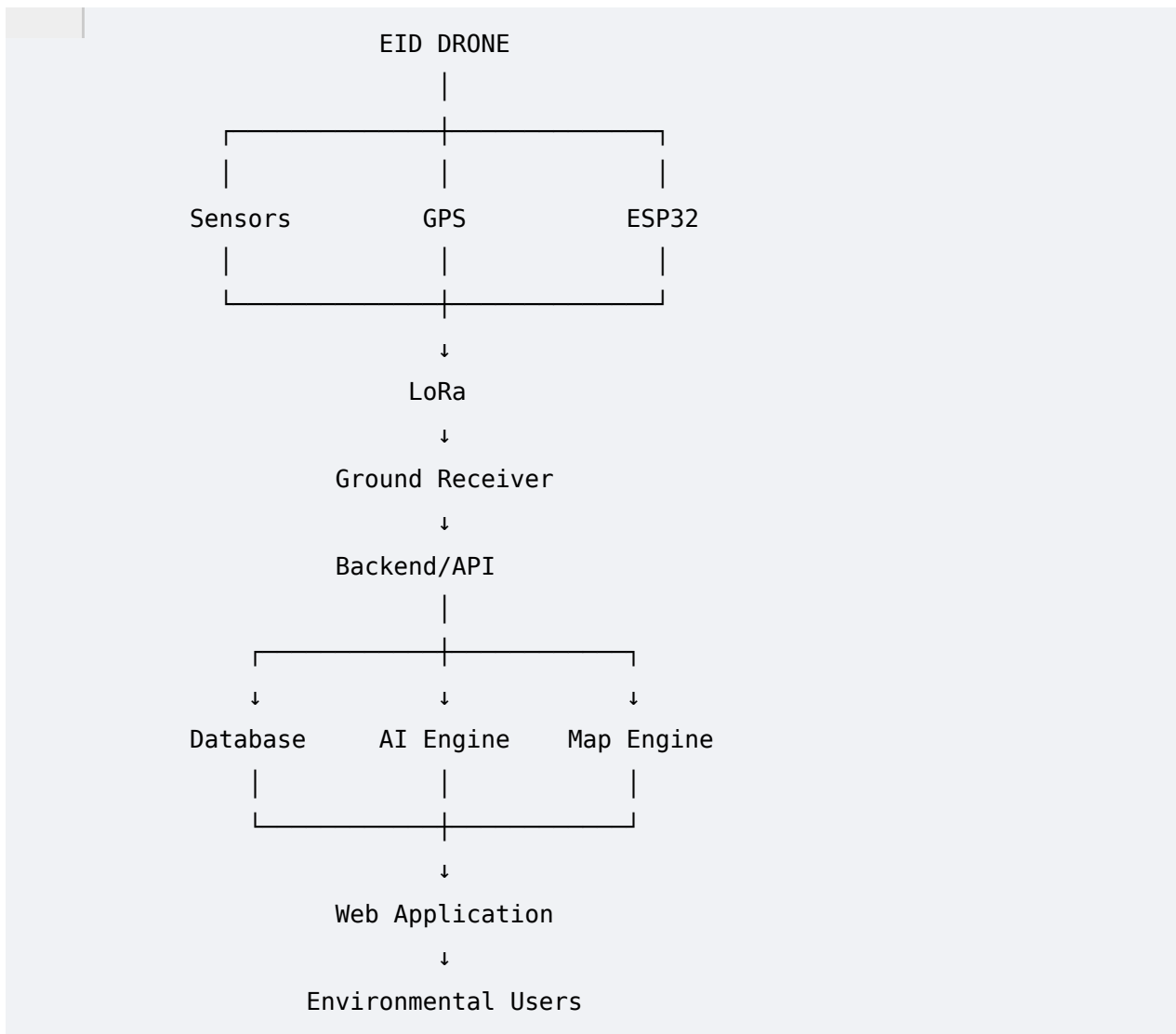
- Monitoring area
- Date/time
- Drone information
- Sensor statistics
- Pollution hotspots
- Historical trends
- AI predictions
- Alerts

- Flight path

Export formats

- CSV
- PDF

21. System Architecture



22. Recommended Technology Stack

Frontend

- React
- TypeScript
- Tailwind CSS
- Leaflet or Mapbox

- Recharts

Backend

Recommended:

- Python FastAPI

Alternative:

- Node.js + Express

Database

Recommended:

- PostgreSQL + PostGIS

Alternative for prototype:

- MongoDB

PostGIS is particularly useful if the system needs advanced geospatial queries.

AI

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost

Future:

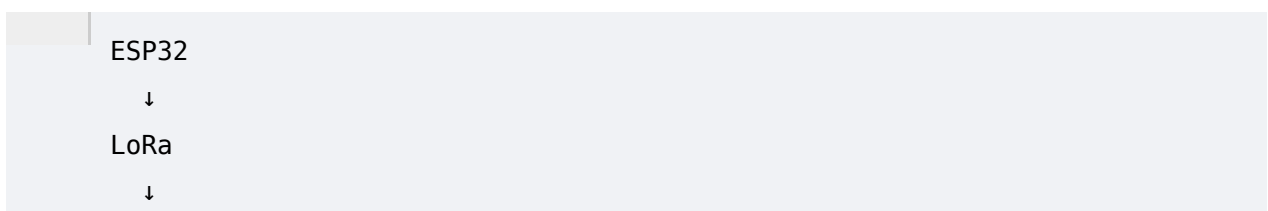
- PyTorch/TensorFlow
- LSTM/GRU

Communication

- LoRa
- REST API
- WebSocket for real-time dashboard updates

23. Real-Time Data Architecture

For live monitoring:



Ground Receiver



Data Parser



Backend



WebSocket



Dashboard

The dashboard should update without requiring the user to manually refresh the page.

24. Non-Functional Requirements

Performance

- Dashboard should load quickly.
- New sensor data should appear with minimal delay.
- Map interactions should remain responsive.

Reliability

- Handle missing sensor data.
- Handle duplicate packets.
- Handle communication interruptions.
- Clearly indicate stale data.

Security

- Secure authentication
- Password hashing
- Role-based authorization
- API authentication
- Input validation
- HTTPS in deployment
- Secure database access

Scalability

The architecture should support:

- Multiple drones
- Multiple monitoring areas
- Additional sensors

- Larger historical datasets
 - Multiple organizations
-

25. Safety and Data Integrity

The platform should clearly distinguish between:

Measured

Direct sensor readings received from the drone.

Processed

Filtered/calibrated/interpolated environmental data.

Predicted

AI-generated estimates.

The UI should use labels such as:

- | LIVE MEASUREMENT
- | PROCESSED DATA
- | AI PREDICTION

This distinction is important for scientific credibility.

26. MVP

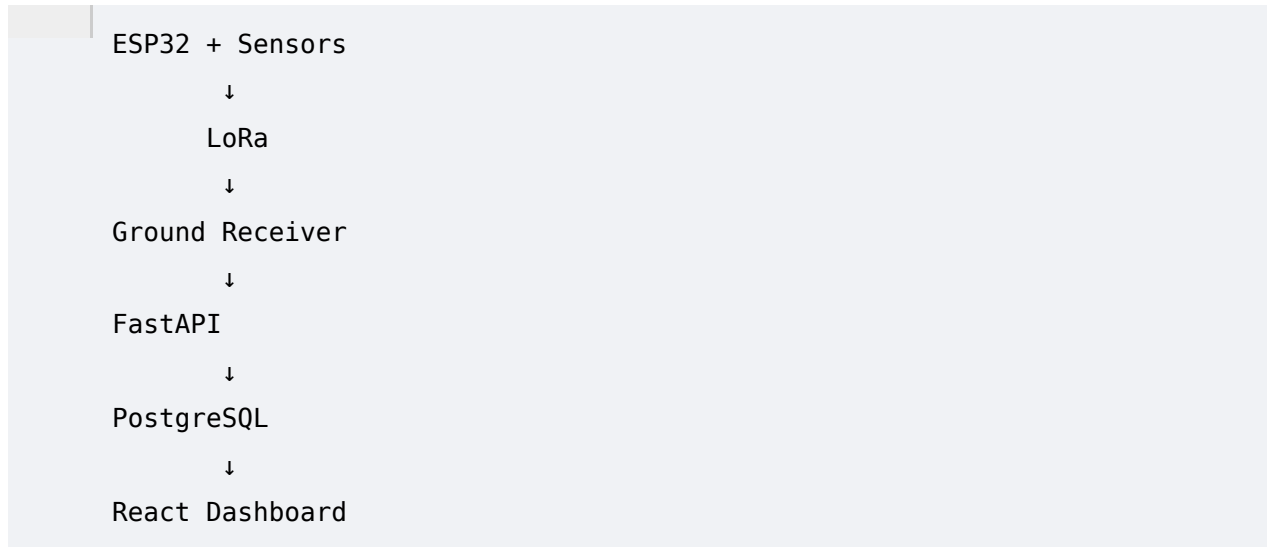
The first version should focus on the smallest useful system.

MVP Features

- Login
- Dashboard
- ESP32 sensor data ingestion
- GPS data
- LoRa data reception
- Live sensor cards
- Interactive map
- Drone location
- Flight path
- Pollution heatmap
- Historical charts

- Basic risk classification
- Alert notifications

MVP Architecture



27. Phase 2

After the MVP:

- AI prediction
- Wind-aware prediction
- Predicted pollution map
- Advanced analytics
- Mission management
- PDF reports
- Email alerts
- Sensor calibration management

28. Phase 3

Future platform:

- Multiple drones
- Cloud infrastructure
- Multi-organization support
- Advanced AI models
- Additional gas/particle sensors
- Thermal camera integration

- Mobile application
 - Smart-city API integration
-

29. Key User Journey

Environmental Analyst

```
graph TD; A[Login] --> B[Dashboard]; B --> C[Select Monitoring Area]; C --> D[View Drone]; D --> E[Monitor Live Sensors]; E --> F[Open Pollution Map]; F --> G[Identify Hotspot]; G --> H[View AI Prediction]; H --> I[Receive Alert]; I --> J[Analyze Historical Data]; J --> K[Generate Report];
```

Login

↓

Dashboard

↓

Select Monitoring Area

↓

View Drone

↓

Monitor Live Sensors

↓

Open Pollution Map

↓

Identify Hotspot

↓

View AI Prediction

↓

Receive Alert

↓

Analyze Historical Data

↓

Generate Report

30. Success Metrics

The platform should be evaluated using measurable technical metrics.

Data

- Percentage of successfully received sensor packets
- Data latency
- Missing-data rate
- GPS tagging success rate

Mapping

- Number of measurements mapped

- Map update latency
- Spatial coverage per mission

AI

- MAE
- RMSE
- R^2
- Precision/recall for risk classification where applicable

Alerts

- Alert generation latency
- False-alert rate
- Missed-alert rate

System

- Dashboard response time
- API availability
- Database reliability

31. Key Design Principles

The website should follow these principles:

1. Data First

The most important information should be visible immediately.

2. Map First

Environmental monitoring is spatial, so the map should be the central visual element.

3. Simple Risk Communication

Use clear Normal/Moderate/High/Critical states.

4. Measured vs Predicted

Always visually distinguish actual sensor readings from AI predictions.

5. Actionable Alerts

An alert should tell the user **what happened, where, when, and how serious it is**.

6. Scientific Credibility

Do not overstate low-cost sensor accuracy or AI prediction certainty.

32. Recommended Homepage

The public homepage should communicate the product in less than 10 seconds.

Hero Section

Environmental Intelligence Drone

Sense. Locate. Map. Predict. Warn.

Subtitle:

An AI-powered environmental monitoring platform that transforms mobile drone sensor data into real-time pollution maps, predictive risk insights, and early-warning intelligence.

CTA buttons:

View Live Dashboard

Explore the Technology

Below Hero

Mobile Sensing



Real-Time Mapping



AI Prediction



Early Warning

33. Final Product Vision

The long-term vision is to create a **mobile environmental intelligence network** where autonomous drones collect spatial environmental data and a centralized platform converts that data into actionable information for municipalities, industries, researchers, and environmental authorities.

The core product philosophy is:

"Environmental pollution is dynamic. Monitoring should be dynamic too."

Final Product Flow

Sense → Locate → Transmit → Process → Map → Predict → Assess → Warn → Act

The EID Web Platform should therefore function not merely as a sensor dashboard, but as a **decision-support platform for proactive environmental monitoring**.